

Where Is the Agent Socially Deficient? Co-Evolving LLM Agents and a Generative Social Sandbox through Critical Moments

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Joint course project proposal: Track A for the Reinforcement Learning course, Track B for the Machine Learning course (UBC). One method, one paper.

1. Problem and insight

Teaching an LLM agent a long-horizon social ability (leading a team, sustaining a relationship) needs scarce real data or full rollouts in a simulator that reproduces the human data-generating process, although most steps are routine. Our prior work found that social outcomes hinge on a few low-entropy *critical moments* [1], and that social behaviour is governed by a few programmable *meta-attributes*, such as bias strengths, set through prompts or steering/task vectors [2,3]. We turn these findings into a training principle: **train only where it matters, only on the meta-attributes that are missing, in a sandbox that itself learns to create those moments**. Both the agent and the sandbox are trainable.

2. Formulation

A long-horizon POMDP has horizon H and trajectories τ . The agent's behaviour is summarized by meta-attributes $m \in \mathbb{R}^d$ (bias strengths, risk attitude, theory-of-mind depth, persona consistency). The target ability has unknown m^* ; the *deficiency* is $\Delta = m^* - m$.

(H1) Temporal sparsity [1]. $G(\tau) = \sum_{t \in C(\tau)} r_t + c$ with $|C(\tau)| \leq K \ll H$ critical steps. **(H2) Decisiveness** [1]. $\mathcal{H}(\pi(\cdot | s)) \leq \delta$ for $s \in C$, so one rollout per critical moment is informative. **(H3) Attribute sparsity** (trait activation [4]). The verifier outcome at a critical moment c is $y_c = \psi(c)^\top \Delta + \xi$ with $\|\psi(c)\|_0 \leq s$: each moment activates only a few meta-attributes. The deficiency is k -sparse, $\|\Delta\|_0 \leq k \ll d$.

Compressed episodes. Under H1, an episode is equivalent to a jump process $(z_0, c_1, a_1, z_1, \dots, c_K, a_K)$. The sandbox's *jump model* $W_\phi(z_j | z_{j-1}, a_{j-1})$ summarizes each routine segment as z_j , routine steps run on a frozen script policy (options [5]), and the agent acts only at the c_j : $O(K)$ agent calls instead of $O(H)$.

3. Method: critical-moment co-evolution

The **agent** A_w is a frozen LLM plus a small controller that sets m_t via steering/task vectors (optionally LoRA), *only at critical moments*. The **sandbox** G_ϕ is an LLM world engine whose actions are moment specifications $c = (\text{context}, \text{trigger}, \text{probed attributes})$ and summaries z . The **verifier** V is calibrated on real human data. Training alternates:

Sense. G_ϕ proposes n moments with sparse designs $\Psi = [\psi(c_1), \dots, \psi(c_n)]^\top$. The deficiency is recovered by $\hat{\Delta} = \arg \min \|\Delta\|_1$ s.t. $\|y - \Psi\Delta\|_1 \leq \eta$. Expander-graph sparse designs satisfy RIP-1, giving ℓ_1 recovery from $n = O(k \log(d/k))$ moments [6]: *trait activation becomes a testable sensing-matrix property*.

Teach. G_ϕ generates moments on $\text{supp}(\hat{\Delta})$ at the agent's frontier, and the agent is updated by policy gradient (GRPO or PPO) on critical steps alone.

Calibrate. A few full rollouts and human data check that compression preserves outcomes; W_ϕ and the moment detector are updated.

The sandbox is trained by RL with a dense reward:

$$r_G = \alpha \underbrace{I(\Delta; y_c | y_{\text{past}})}_{\text{sense}} + \beta \underbrace{[J_c(w_{t+1}) - J_c(w_t)]}_{\text{teach}} - \lambda \underbrace{D(P_{\text{full}}(G) \| P_{\text{comp}}(G))}_{\text{faithfulness}}.$$

Target result (to prove, extending Thm. 1 of [1]): if each jump has outcome error at most ε_W , then $|J_{\text{full}}(\pi) - J_{\text{comp}}(\pi)| \leq K(R_{\text{max}}\varepsilon_W + L\delta)$. Compressed training is provably close to full training at a fraction of the cost.

4. What is new

vs. PAIRED/PLR/ACCEL [7,8]: they generate full environments, estimate regret per environment with full rollouts, and give the generator a sparse regret reward; we generate compressed episodes of critical moments, separate measuring from teaching, and use a dense reward with a faithfulness term. **vs. SOTOPIA- π** [9]: whole-dialogue training, no diagnosis. **vs. jumpy world models/options**: our jump points are chosen by a trainable sandbox. **vs. synthetic-population benchmarks** [10]: they diagnose; we diagnose and repair.

5. Two tracks

Track A: algorithm (RL)	A synthetic <i>sparse-critical POMDP</i> suite with controllable (H, K, d, k, s) and known Δ , to test H1–H3, the recovery bound and the compression bound. Long-horizon text games (TextWorld, ALFWorld) with an LLM jump model. Baselines: full-rollout PPO/GRPO, DR, PAIRED, Robust PLR. Metrics: recovery error vs. n , return vs. agent calls, faithfulness. No social content needed.
Track B: social sandbox (ML)	6–10 meta-attributes from [2] as steering/task vectors; a moment detector trained by counterfactual turn resampling, $u_t = \mathbb{E}[V \text{resample } t] - V$; a sandbox of script graphs, perturbations and NPC meta-attributes (visualized by our town demo). Evaluation: SOTOPIA(-hard) goal completion; faithfulness on the speed-dating and divorce data of [1]; human ratings on a subset.

Questions. Q1: do H1–H3 hold (sparsity of Ψ , concentration of advantage)? Q2: at equal agent calls, does co-evolution beat full-rollout training and PAIRED/PLR? Q3: is $\hat{\Delta}$ interpretable and stable across seeds? **Risks.** If H3 fails, use dense designs with LASSO; for a noisy verifier, average judges and check against human ratings. **Plan.** Wk 1–3 synthetic suite, attribute vectors; 4–7 Sense/Teach (A), detector (B); 8–11 full co-evolution; 12–13 write-up, demo.

[1] Shang, Yan, Liu. Love First, Know Later: Persona-Based Romantic Compatibility through LLM Text World Engines. NeurIPS 2025 Workshop on LLM Persona Modeling. [2] Liu, Shang, Jin. Programmable Cognitive Bias in Social Agents. arXiv:2509.13588 (CHI 2026). [3] Liu et al. Exploring Prosocial Irrationality for LLM Agents: A Social Cognition View. arXiv:2405.14744. [4] Tett, Burnett. A personality trait-based interactionist model of job performance. J. Appl. Psych. 2003. [5] Sutton, Precup, Singh. Between MDPs and Semi-MDPs. AIJ 1999. [6] Berinde, Gilbert, Indyk, Karloff, Strauss. Combining geometry and combinatorics: a unified approach to sparse signal recovery. Allerton 2008. [7] Dennis et al. PAIRED. NeurIPS 2020. [8] Jiang et al. Robust PLR. NeurIPS 2021; Parker-Holder et al. ACCEL. ICML 2022. [9] Zhou et al. SOTOPIA. ICLR 2024; Wang et al. SOTOPIA- π . ACL 2024. [10] Chidichimo et al. Artificial Societies Benchmark. arXiv:2609.30030, 2026.